7:07 186 B/s 60

Quiz
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BC180402531: MARYAM SIMAB

Time Left 76 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 5 of 10 (Start time: 07:06:48 AM, 02 March 2022)

Total Marks: 1

Which of the following is a condition of an inner product space?

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Select the correct option

Reload Math Equations

- $\langle \alpha x, y \rangle = \langle x, \alpha y \rangle$
- $\langle x, y \rangle = \overline{\langle x, y \rangle}$
- $\langle \alpha x + y, z \rangle = \langle x, z \rangle + \alpha \langle y, z \rangle$
- $\langle x, x \rangle \geq 0$

Click to Save Answer & Move to Next Question

R

Mod 83

11:51

Quiz
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MC200402300: ASIF ALI

Time Left 36 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 3 of 10 (Start time: 11:50:29 AM, 01 March 2022)

Total Marks: 1

Since a bounded linear operator T from the normed space X to normed space Y is defined and given as;

$\forall x \in D(T)\ \exists\ k > 0$, such that $\|Tx\| \leq k\|x\|$, then the maximum value of k is_____

Select the correct option

Reload Math Equations

- $\sup_{x \in D(T), x \neq 0} \frac{\|Tx\|}{\|x\|}$
- $\inf_{x \in D(T), x \neq 0} \frac{\|Tx\|}{\|x\|}$
- not defined
- any arbitrary non negative real number

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11:55

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MC200402300: ASIF ALI

Time Left 65 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 5 of 10 (Start time: 11:54:57 AM, 01 March 2022)

Total Marks: 1

Which of the following is a condition of an inner product space?

Select the correct option

Reload Math Equations

- $\langle x, x \rangle \geq 0$
- $\langle \alpha x + y, z \rangle = \langle x, z \rangle + \alpha \langle y, z \rangle$
- $\langle \alpha x, y \rangle = \langle x, \alpha y \rangle$
- $\langle x, y \rangle = \overline{\langle x, y \rangle}$

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11:53

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MC200402300: ASIF ALI

Time Left 13 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 4 of 10 (Start time: 11:51:30 AM, 01 March 2022)

Total Marks: 1

On a normed space X of all n degree polynomials defined on $[-1,1]$, if D is a differential linear operator defined and given as; $D(x(t)) = \left\{\frac{d}{dt}x(t) : x(t) \in X, -1 \leq t \leq 1\right\}$, then $\forall x \in X \; \exists k > 0$ such that $\|Dx\| \leq k\|x\| \Longrightarrow$

Select the correct option

Reload Math Equations

- D is a linear bounded
- D is non-linear unbounded
- D is linear unbounded
- D is a non-linear bounded

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7:04

Quiz
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BC180402531: MARYAM SIMAB

Time Left 32 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 2 of 10 (Start time: 07:03:17 AM, 02 March 2022)

Total Marks: 1

Null space is also a

Select the correct option

- Vector space.
- Linear functional.
- Canonical mapping.
- Metric space.

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7:09

390 B/s 60

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BC180402531: MARYAM SIMAB

Time Left 65 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 7 of 10 (Start time: 07:08:37 AM, 02 March 2022)

Total Marks: 1

Dot product is a

Select the correct option

- Function.
- Mapping.
- Functional.
- operator.

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7:08

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BC180402531: MARYAM SIMAB

Time Left 13 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 6 of 10 (Start time: 07:07:10 AM, 02 March 2022)

Total Marks: 1

For n dimensional vector space and its dual space we have

Select the correct option

Reload Math Equations

- $D(X) = X^* = n$
- $\dim(X) \subset \dim(X^*)$
- $R(X) = X^* = n$
- $\dim(X) = \dim(X^*) = n$

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7:03

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BC180402531: MARYAM SIMAB

Time Left 18 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 1 of 10 (Start time: 07:01:58 AM, 02 March 2022)

Total Marks: 1

The mapping $T : V \to V$ defined by $T(v) = a + v$, where $v \in V$ is linear if

Select the correct option

Reload Math Equations

- $a = 0$
- $a \neq 0$

Click to Save Answer & Move to Next Question

7:10

Quiz
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BC180402531: MARYAM SIMAB

Time Left 63 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 9 of 10 (Start time: 07:10:21 AM, 02 March 2022)

Total Marks: 1

Every complete inner product space is ---------.

Select the correct option

Reload Math Equations

- Complex space
- Hilbert space
- Banach space
- Euclidean space

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3:36 P

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MC200402300: ASIF ALI

Time Left 27 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 6 of 10 (Start time: 03:35:29 PM, 01 March 2022)

Total Marks: 1

On a normed space X of all polynomials of form $x(t) = t^{n+1}, n \in \mathbb{N}$ defined on $[-1,1]$,if D is a differential linear operator defined and given as;

$D(x(t)) = \left\{\frac{d}{dt}x(t) : x(t) \in P[-1,1], -1 \leq t \leq 1\right\}$, then $\|D\| =$_____.

Select the correct option

Reload Math Equations

- ( ) $n+1$
- ( ) 0
- ( ) 1
- (•) n

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7:10

Quiz
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BC180402531: MARYAM SIMAB

Time Left 37 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 8 of 10 (Start time: 07:09:14 AM, 02 March 2022)

Total Marks: 1

Which of the following an example of Linear Functional?

Select the correct option

Reload Math Equations

- (●) the integral operator $I : c[0,1] \to \mathbb{R}$, on the space of all contnuous functions on $[0,1]$ defined by $I(f) = \int_0^1 f(t)\,dt$
- ( ) the derivative operator on space of all real polynomials

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7:11

6.7 K/s 59

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BC180402531: MARYAM SIMAB

Time Left 65 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 10 of 10 (Start time: 07:10:56 AM, 02 March 2022)

Total Marks: 1

If a normed space X is finite dimensional, then every linear operator on X is

Select the correct option

Reload Math Equations

- Continuous.
- Differentiable.
- Bounded.
- Integrable.

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3:36 P

3.4 K/s 76

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MC200402300: ASIF ALI

Time Left 27 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 6 of 10 (Start time: 03:35:29 PM, 01 March 2022)

Total Marks: 1

On a normed space X of all polynomials of form $x(t) = t^{n-1}, n \in \mathbb{N}$ defined on $[-1, 1]$, if D is a differential linear operator defined and given as;

$D(x(t)) = \left\{ \frac{d}{dt} x(t) : x(t) \in P[-1, 1], -1 \leq t \leq 1 \right\}$, then $\|D\| =$_____

Select the correct option

Reload Math Equations

- $n+1$
- 0
- 1
- n

Click to Save Answer & Move to Next Question

6:54

Quiz
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MC200402300: ASIF ALI

Time Left 15 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 7 of 10 (Start time: 06:53:03 AM, 02 March 2022)

Total Marks: 1

If a sequence $x_n \to x$ in a normed space X, then for a bounded linear operator T on X, then

Select the correct option

Reload Math Equations

- $Tx_n \to Tx$
- $Tx_n \to Tx$

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6:56

115 B/s 62

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MC200402300: ASIF ALI

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Time Left 62 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 9 of 10 (Start time: 06:55:46 AM, 02 March 2022)

Total Marks: 1

A hilbert space is a /an.....

Select the correct option

- incomplete norm space
- incomplete inner product space
- complete norm space
- complete inner product space

R

Click to Save Answer & Move to Next Question

7:10

Quiz
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BC180402531: MARYAM SIMAB

Time Left 63 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 9 of 10 (Start time: 07:10:21 AM, 02 March 2022)

Total Marks: 1

Every complete inner product space is ----------.

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Select the correct option

Reload Math Equations

- Complex space
- Hilbert space
- Banach space
- Euclidean space

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7:11

6.7 K/s 59

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BC180402531: MARYAM SIMAB

Time Left 65 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 10 of 10 (Start time: 07:10:56 AM, 02 March 2022)

Total Marks: 1

If a normed space X is finite dimensional, then every linear operator on X is

Select the correct option

Reload Math Equations

- Continuous.
- Differentiable.
- Bounded.
- Integrable.

Click to Save Answer & Move to Next Question

6:56 186 B/s 62

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MC200402300: ASIF ALI

Time Left 57 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 11:48 AM, 01 March 2022

Question # 10 of 10 (Start time: 06:56:24 AM, 02 March 2022)

Total Marks: 1

In an Inner Product space say X, if the sequences $\{x_n\}$ and $\{y_n\}$ are Cauchy, then $\langle x_n, y_n \rangle$ is ----------

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Select the correct option

Reload Math Equations

- necessarily a Cauchy Sequence in X
- not necessarily a Cauchy Sequence in F
- not necessarily a Cauchy Sequence in X
- necessarily a Cauchy Sequence in F

Click to Save Answer & Move to Next Question

Mid 90 R

7:09

390 B/s 60

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BC180402531: MARYAM SIMAB

Time Left 65 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 7 of 10 (Start time: 07:08:37 AM, 02 March 2022)

Total Marks: 1

Dot product is a

Select the correct option

- Function.
- Mapping.
- Functional.
- operator.

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7:06

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BC180402531: MARYAM SIMAB

Time Left 67 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 4 of 10 (Start time: 07:05:54 AM, 02 March 2022)

Total Marks: 1

For an inner product space $\langle x+y,z\rangle =$ _____

Select the correct option

- $\langle x,z\rangle \cdot \langle y,z\rangle$
- $\langle x,z\rangle + \langle y,z\rangle$
- $\langle x,z\rangle - \langle y,z\rangle$
- None of these

Click to Save Answer & Move to Next Question

R

Mod 83

7:08

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BC180402531: MARYAM SIMAB

Time Left 13 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

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Quiz Start Time: 07:01 AM, 02 March 2022

Question # 6 of 10 (Start time: 07:07:10 AM, 02 March 2022)

Total Marks: 1

For n dimensional vector space and its dual space we have

Select the correct option

Reload Math Equations

- $D(X) = X^* = n$
- $\dim(X) \subset \dim(X^*)$
- $R(X) = X^* = n$
- $\dim(X) = \dim(X^*) = n$

Click to Save Answer & Move to Next Question

✓ R

Mod 75

7:10

0 B/s 60

Quiz
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BC180402531: MARYAM SIMAB

Time Left 37 sec(s)

MTH641 - Functional Analysis (Quiz NO.03)

Quiz Start Time: 07:01 AM, 02 March 2022

Question # 8 of 10 (Start time: 07:09:14 AM, 02 March 2022)

Total Marks: 1

Which of the following an example of Linear Functional?

Select the correct option

Reload Math Equations

- (●) the integral operator $I : c[0,1] \to \mathbb{R}$, on the space of all contnuous functions on $[0,1]$ defi... $I(f) = \int_0^1 f(t)\,dt$
- ( ) the derivative operator on space of all real polynomials

Click to Save Answer & Move to Next Question